

LO9 - 21/03/2024

goal: We are going to use ORTHOGONAL MATCHING PURSUIT and LEAST SQUARE ORTHONORMAL MATCHING PURSUIT in order to implement the denoising procedure. So, let's see OMP DENOISING:

Let's consider the following setting:

1. $Y \in \mathbb{R}^{K \times C}$, $S = Y + \eta$, $\eta \sim N(0, \sigma^2)$
2. the algorithm that we have seen are based on local processing:
 - a. Extract from S a patch $s \in \mathbb{R}^{p \times p}$ (or a vector $s \in \mathbb{R}^H$, $H = p^2$)
 - b. Considering DCT denoising, we are going to process all the patches independently by:
 - compute the representation $x = D^T s$, where $D \in \mathbb{R}^{H \times H}$ orthonormal $D^T D = D D^T = I_H$ (ANALYSIS)
 - perform the hard thresholding $\hat{x} = HT_\tau(x)$ in order to enforce sparsity (HARD THRESHOLDING)
 - compute the estimate of the patch $\hat{s} = D \hat{x}$ (SYNTHESIS)
 - c. perform the aggregation in order to reconstruct the image $\hat{S}$

In the highlighted steps, we are computing the sparse representation of our signal s w.r.t D . In fact, $\|x\|_0$ is small. Note that we are able to compute those steps because we are using an orthonormal basis D .

In the case in which $D \in \mathbb{R}^{M \times N}$ ($M < N$), we cannot compute the sparse representation of s w.r.t D using the previously mentioned approach. In this case, we can rely on the orthogonal matching pursuit algorithms. We'll perform the following steps:

1. Extract from S a patch $s \in \mathbb{R}^{p \times p}$ (or a vector $s \in \mathbb{R}^H$, $H = p^2$)
2. Compute $\hat{x}$ using $OMP(s, D, L, \tau) \rightarrow s$: patch, D : dictionary, L, τ : stopping criteria
3. compute the estimated patch $\hat{s} = D \hat{x}$

Let's see more in depth the OMP algorithm:

Setting: $D \in \mathbb{R}^{M \times N}$ with $M < N$, $s \in \mathbb{R}^M$, $x \in \mathbb{R}^N$

Goal: $s \approx Dx$ s.t $\|x\|_0$ small

Init: $x^{(0)} = 0$, $r^{(0)} = s - Dx^{(0)}$, $W^{(0)} = \{\emptyset\}$, $l = 1$
ACTIVE SET

while stopping criteria: $\rightarrow \|r^{(l)}\| > \tau$ and $\|\hat{x}^{(l)}\|_0 < L$

sweep step: $k = 1 \dots N$:

...

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To perform the denoising we have to set the sparsity level and the threshold τ . We have seen that setting a sparsity level L is not optimal because we do not know how many atoms we need in order to reconstruct the patch. Each patch will need its own L .

idea: set in our OMP $L=M$. The latter means that we are not imposing any constraint on $\|\hat{x}\|_0$.

The only condition that we are going to look is actually $\|r^{(i)}\| > \tau$. Hence, we need to set τ .

When it comes to DCT denoising, we said that $\tau = 3 \cdot \sigma$.

Let's now consider a noise-free patch and assume that y can be written as $y = Dx$ where

x is sparse. We'll have that:

$S = Dx + \eta \xrightarrow{\eta_i \sim \mathcal{N}(0, \sigma^2)} r = S - Dx \longrightarrow \|r\|_2^2$. Since the latter is a random variable we can compute

its expected value: $E[\|r\|_2^2] = E[\|S - Dx\|_2^2] = E[\|\eta\|_2^2] = E[\sum_{i=1}^M \eta_i^2] = \sum_{i=1}^M E[\eta_i^2] = M \cdot \sigma^2$

Based on that result, we expect τ to be more or less proportional to $\sqrt{M} \cdot \sigma$. manually tuned

So, in our stopping criteria we'll have that: $\|\hat{x}\|_0 < L$ and $\|S - D\hat{x}\| > \tau = \lambda \sqrt{M} \cdot \sigma$ (↻)

Let's write down the denoising procedure:

1. start from the image S

2. for $i = 0, \dots, r-p$

3. for $j = 0, \dots, c-p$

4. extract patch s

5. $\hat{x} = \text{OMP}(s, D, M, c) \xrightarrow{L=M}$

6. estimate of the patch $\hat{y} = D\hat{x}$

7. image reconstruction $\hat{Y}[i:i+p, j:j+p] += \hat{y} \cdot \textcircled{w} \longrightarrow$ the uniform weight is the best choice when we are using OMP. In fact, since we need $\text{var}(\hat{y})$ in order to compute the weights and $\text{var}(\hat{y})$ is not proportional to $\|\hat{x}\|_0$, we have to use the uniform weight

8. update the weights $W[i:i+p, j:j+p] += w$

9. normalize $\hat{Y} = \hat{Y} / W$

observation: In our case we have that: $s \sim \mathcal{N}(Dx, \sigma^2 I) \longrightarrow x = D^T s \sim \mathcal{N}(x, \sigma^2 I)$

if we compute the expected value we'll obtain: $E(x) = E(D^T s) = D^T E(s) = D^T Dx = x$

We can also compute the variance of $D^T s$: $\text{Var}(D^T s) = \sigma^2 D^T I D = \sigma^2 I$